A Sociotechnical Systems Primer for Systems Engineers: Definitions and mapping

Acknowledgements

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Let people know this is the first volume of a series

Volume 2 would be the in-practice using Shams's paper

Volume 3 could then be case studies

Follow up volumes could deal with the short pieces that arise out of Figure 2

Look for a Fellow who might provide a welcome

SECTION 1: INTRODUCTION

Sociotechnical systems are fundamental to systems engineering. As a simple example, Figure 1 provides a generic outline of the life-cycle stages in a typical systems engineering project, every one of which involves some kind of sociotechnical system. During the concept phase, a problem space needs to be defined, usually in a sociotechnical process that involves at least one engineer working with at least one stakeholder. That simple interaction creates a social system that can be optimized using tools from the social sciences. At the same time, the social interaction in the concept phase will necessarily involve a technical apparatus of some kind. And these parameters of people and technology, central to the definition of sociotechnical systems, run through every stage in the life-cycle process.

TABLE 3.1 Generic life cycle stages, their purposes, and decision gate options

Life cycle stages	Purpose	Decision gates
Concept	Define problem space	Decision options
	1. Exploratory research	• Proceed with next stage
	2. Concept selection	• Proceed and respond to action items
	Characterize solution space	• Continue this stage
	Identify stakeholders' needs	• Return to preceding stage
	Explore ideas and technologies	• Put a hold on project activity
	Refine stakeholders' needs	• Terminate project
	Explore feasible concepts	
	Propose viable solutions	
Development	Define/refine system requirements	
	Create solution description—architecture and design	
	Implement initial system	
	Integrate, verify, and validate system	
Production	Produce systems	
	Inspect and verify	
Utilization	Operate system to satisfy users' needs	
Support	Provide sustained system capability	
Retirement	Store, archive, or dispose of the system	

Figure 1. Generic life cycle stages.

Even these simplest sociotechnical systems can be optimized using combinations of ideas from systems engineering and social sciences, and that realization is at the heart of social systems/sociotechnical systems engineering (see Garcia-Diaz and Olaya, 2018).

The concept of the engineering of sociotechnical systems is not new. The expression social systems engineering can be traced back to the mid-1970s, and the Proceedings of the IEEE published a special issue on it in 1975 (Chen et al). Here the authors focused on using systems engineering concepts to work on social problems. Systems engineers continue to work on problems where a human/social variable is a primary aspect of the system problem under consideration, and there is always a technological component involved that makes the system functionally sociotechnical.

Recognizing the sociotechnical nature of systems engineering work thus has the potential to provide systems engineers with a strong foundation for understanding, coping with and developing the social side of technical systems. The large and developing literature on sociotechnical systems can guide systems engineering in this aspect of its practice.

This paper has been developed from the basis of a systematic literature review to give systems engineers an overview of sociotechnical systems in the systems engineering literature. The paper is intended to provide a necessary foundation for the development of sociotechnical systems engineering practice by creating a taxonomy of definitions of sociotechnical systems.

Section II of this paper defines sociotechnical systems at its most elementary level and provides a list of papers for further reading.

Section III of this paper shows how definitions of sociotechnical systems change and evolve as the problem spaces of systems of interest become more complex. The addition of complexity in the problem spaces under consideration forces systems engineers to use new language in discussion of the problems. Section III provides a mapping of the way this added complexity pushes systems engineers to bring ideas from a wide variety of academic fields into their engineering work. The section ends with a list of papers for further reading in this more complex version of sociotechnical systems. Section IV provides a concept map that summarizes our findings.

SECTION II. The Generic/Simplest Definition of Sociotechnical Systems

The generic definition used by systems engineers in discussion of their practice is that sociotechnical systems involve humans interacting with technology. In the most basic discussions of sociotechnical systems, the definition of the term is unstated. For example:

> "Finally, [this paper] takes the lessons learned from the project, and the analysis, to reflect on how SE needs to change to improve its capacity to guide and manage interventions in complex, sociotechnical systems where the concept is less appropriate for its current methodology" (Rodenas et al. 2021).

It is common in scientific and engineering writing to include conceptual definitions of terms that are outside the common core language of the community being addressed by the paper, so it can be assumed that authors who do not add conceptual definitions of sociotechnical systems have a common meaning for their community of practice. We hypothesize that the assumed definition is the relatively simple, generic definition stated above.

Whenever the definition becomes more complex, conceptual definitions of additional tend to be provided, as you will see in later sections of the primer. However, the literature pointed out quite a few examples of papers where the generic definition is given a slightly different or more refined definition. For example:

> "In complex sociotechnical systems, cognitive and social humans use technology to make sense of situations when making decisions" (Oosthuizen and Pretorius, 2016).

> "A socio-technical system is a collection of stakeholders, their networks, practices, and knowledge; the technologies they use; their collective representations; and the standards and rules they adopt" (Yang et al 2021).

Note that even at this simplest level, the system of interest can involve quite a few elements that have potential complexity, and this complexity offer up the possibility that systems engineers might need tools from other fields (for example the social sciences) to optimize their working methods. In the example from Yang et al, a collection of stakeholders has a network structure that might need analysis (using elements of network theory); their collective practices and knowledge base might need to be understood (bringing in techniques from the social sciences), the standards and rules need understanding as well (using techniques from the social sciences); and, finally, there are the technologies at the heart of the system that are the typical foci of SE.

Hence, the use of the term *sociotechnical system* varies in definitional specificity from generic to technically more complex. On the one hand, the term can refer generically to any system that has an intersection of social and technical systems, which is the definition used in the Systems Engineering Book of Knowledge (SEBoK).[2] On the other hand, there are well-established theoretical concepts, referred to as sociotechnical systems, in a variety of disciplines (for example, the social science disciplines[6]).

What we see through the literature upon which this primer is based is that, within systems engineering, the use of the term *sociotechnical system* is increasing, and there are a range of definitions of this term in the systems engineering literature. These different definitions are based on a varied literature base that comes from several different engineering disciplines, often reflecting the domain specific application of systems engineering. Because the literature on sociotechnical systems in systems engineering is so scattered, there is a need for a common understanding of how this term is being used and applied in systems engineering literature, especially when the term includes aspects that move beyond the simple definition stated above.

SECTION III. More complex definitions of sociotechnical systems

Within the systems engineering literature on sociotechnical systems, as the definition becomes more complex, researchers discussing the term tend to become more explicit in the definition they use, adding elements to it that are particular to the problems they encounter. The following are the most common aspects of sociotechnical systems that systems engineers refer to beyond the simple definition. For a deeper analysis of these additional elements, we refer you to Polojärvi et al 2023. The list is presented here with a nested structure. The first bullet point has a subset of other bulleted points.

- Sociotechnical systems are situated within an external environment.
    - Because of this environmental context, sociotechnical systems are open systems.

    - Sociotechnical systems often include hierarchies of scale within the system of systems of which any given sociotechnical systems are a part (for example, sociotechnical systems within socioecological systems)
    - These socio-environmentally situated sociotechnical systems can interact with other sociotechnical systems.
- Sociotechnical systems are complex systems
    - Because these systems are complex, they can involve goal driven behavior
    - The difficult to predict interactions between elements of these systems can lead to emergent behavior that is unpredictable and difficult to uncontrol.
    - Sociotechnical systems contain many possible interactions between agents, and elements of these systems can be:
        - Adaptive (the agents are able to change their behavior in response to the outputs of interactions in the system)
        - Evolutionary (the adaptations can potentially lead to transformations in the state of the systems)
- The interacting agents in a sociotechnical system can have a variety of roles and qualities within the system of which they are a part:
    - They self-organize in response to system rules
    - They have goals
    - They are adaptive
    - They can be empowered
    - They can be hybrid in nature (serving multiple functions)
    - They can compete
- Sociotechnical systems can involve big data. This is defined by Dragoicea et al.[68]: "socio-technical systems are complex systems in which big data is involved, in volume, diversity of sources, variety, velocity and veracity.[56]
- Sociotechnical systems can be defined as a dynamic configuration of resources. This is also defined by Dragoicea et al.[68]: "according to the service dominant logic,[57] socio-technical systems can be also approached as dynamic configurations of resources where value co-creation processes can be highlighted.[58, 59]

SECTION IV: Concept mapping of sociotechnical systems

The definition of *sociotechnical system* has origins in systems science for most systems engineering documents or sociotechnical systems theory for a sub-set of documents from the ergonomics and human systems integration systems engineering specialties.

These findings have been summarized in a concept map (see Figure 2).

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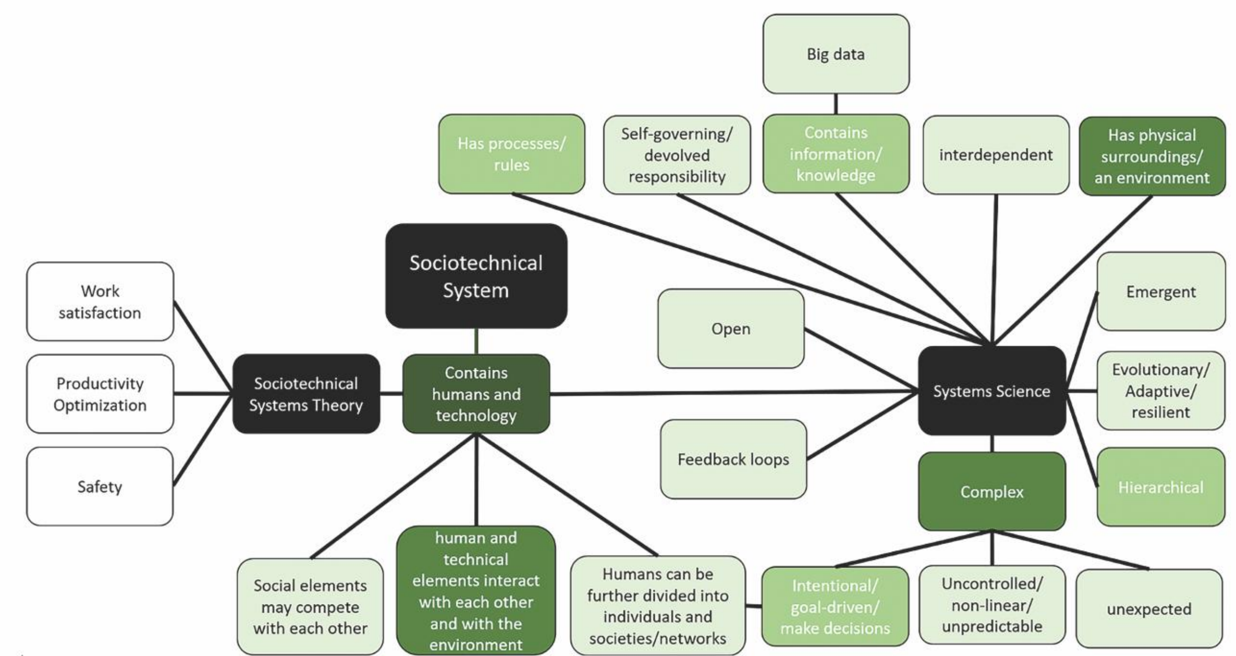

Figure 2. Sociotechnical System Concept Map

These two origin theories are used in the concept map to structure the concepts. The shade of green indicates the frequency of the concept in the systems science-based definitions of *sociotechnical system*. The darker the green the more frequently that concept was used in the definition of *sociotechnical system*.

END

Works cited:

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I. : How sociotechnical systems are important to systems engineering: and structure of the paper

II What is socio-technical Systems

Give basic definition and the standard one and outline of rest of document and

Socio techincal systems play importnat role in systems engineering. For systems engineers we define socio-technical system as " "

Concept map

This primer is based on a systematic literature review of sociotechnical systems in systems engineering https://doi.org/10.1002/sys.21664 and needs to be edited and summarized for the primer, but it serves as a starting point.

Define method to show rigour

Systems engineers use the term *sociotechnical system* in academic literature and in their practice. Sociotechnical systems are gaining more attention as systems engineers aspire to address the social elements of their systems engineering practice as well as societal challenges, such as those included in the International Council on Systems Engineering (INCOSE) Vision 2035.[1] Even though there is a basic working definition of *sociotechnical system* in the Systems Engineering Book of Knowledge (SEBoK),[2] use of the term varies in systems engineering literature depending on application domain. As systems engineering research and practice venture into social domains, it is critical that systems engineers have a shared understanding of terms they use as a foundation of knowledge and practice. The term *sociotechnical system* is used in many different disciplines including systems engineering. The use of the term varies in definitional specificity from generic to technically more complex. On the one hand, the term can refer generically to any system that has an intersection of social and technical systems, which is the definition used in the Systems Engineering Book of Knowledge (SEBoK).[2] On the other hand, there are well-established theoretical concepts, referred to as sociotechnical systems, in a variety of disciplines (for example, the social science disciplines[6]). Within systems engineering, the use of the term *sociotechnical system* is increasing, and there are a range of definitions of this term in the systems engineering literature. These different definitions are based on a varied literature base that comes from several different engineering disciplines, often reflecting the domain specific application of systems engineering. Because the literature on sociotechnical systems in systems engineering is so scattered, there is a need for a common understanding of how this term is being used and applied in systems engineering literature.

II.

Sociotechnical Systems: A Definition (based on literature review https://doi.org/10.1002/sys.21664)

Our systematic literature review resulted in our recognition of two main findings. First, there is a clear pattern within the definitions themselves. The vast majority of the papers that define the term *sociotechnical system* either formally or by context share a similar and fairly simple definition, that sociotechnical systems involve humans interacting with technology. However, a few of the definitions are much more specific about the potential complexity of sociotechnical systems, including concepts such as emergence and scalar hierarchies. In the first part of our discussion, we will share some of these definitions explicitly, so our readers can get an immersive sense of the range of definitional possibility in the current state of the art of the use of the term *sociotechnical system* in the systems engineering literature. This section will be organized to show the increasing complexity of the definitions provided. The second main finding is that two theoretical points of departure emerged from our review of the papers. The first used the sociotechnical systems theory approach (in the tradition of Rasmussen)[8,9]and is found in papers from ergonomics and some of the papers from human factors and safety. This sociotechnical systems theory perspective was a minority approach in the papers in our selection. We will discuss this finding last and will focus first on the second theoretical perspective based on systems science.

Analysis of definitional complexity (based on discussion in https://doi.org/10.1002/sys.21664)

The generic definition we observed in the set of papers that we included in our analysis is quite simple: Sociotechnical systems involve humans interacting with technology. For example:

> "A socio-technical system is a collection of stakeholders, their networks, practices, and knowledge; the technologies they use; their collective representations; and the standards and rules they adopt".[10]

Based on this generic definition, we were able to identify three groups of papers with distinct definitional differences from zero definition provided to reaffirmation of the generic definition to a more complex and detailed definition.

Group 1

Twenty-three of the papers included in our search used the term *sociotechnical system* without any attempt at definition at all. For example:

"Finally, [this paper] takes the lessons learned from the project, and the analysis, to reflect on how SE needs to change to improve its capacity to guide and manage interventions in complex, sociotechnical systems where the concept is less appropriate for its current methodology".[11]

> "Since it is challenging to perform empirical studies directly on real organizations at scale, some researchers in systems engineering and design have begun relying on abstracted model worlds that aim to be representative of the reference socio-technical system, but only preserve some aspects of it".[12]

Given that it is a common practice in scientific and engineering writing to include conceptual definitions of terms that are outside the common core language of the discourse community being addressed by the paper, it could be assumed that the authors in Group 1 believe the term to have a common meaning for their community of practice. We hypothesize that the assumed definition is the relatively simple, generic definition stated above. As you will see in the discussion below, when the definition becomes more complex, conceptual definitions are provided. In the discussion below, you will see mention of "environment", and this term has potentially interesting complications. In some cases, the term environment is clearly the immediate physical environment of the humans engaged with their technology, and in others it is a more complex recognition of the human-technology-work environment system embedded in the larger-scale natural environment. We see this latter point as an additional level of complexity.

Group 2

A second set of 23 papers defined the term generically but purely through context. Here is a random, chronological sampling of the generic definitions in table form so you can see the way the definitions appear.

Table 1. Examples of generic contextual definitions of *sociotechnical system*.

Relevant quotations	**Publication**
"...sociotechnical domains where humans cooperate with computerized systems to ensure safe and efficient system behavior."	Borst et al. (2015)[13]
"In complex sociotechnical systems, cognitive and social humans use technology to make sense of situations when making decisions."	Oosthuizen and Pretorius (2016)[14]
"Sociotechnical enterprises represent a microsociological construct."	Donaldson (2017)[3]

"sociotechnical systems... human behavior and organizational context play important roles"	Szajnfarber and Gralla (2017)[15]
"socio-technical problem (i.e., solutions are focused on humans interacting with technology)"	Bennett et al (2018)[16]
"sociotechnical system, highlighting the interaction between pilots, VTS operators and their work systems and environment"	De Vries et al. (2019)[17]
"Socio-technical components within the work system include Person factors (e.g., providers and patients), Task factors (e.g., variety and complexity of tasks), Technology factors (e.g., health information technology [HIT]),[18] Organization factors (e.g., institutional mandate for communicating critical results to ordering provider,[19] care coordination), and Environmental factors (e.g., physical layout, noise)."	Lacson et al. (2019)[20]
"a socio-technical view of man–machine systems"	Lundberg et al. (2020)[21]

Please note that, of the papers in *Group 2*, two (De Vries et al. 2019[17]; Lacson et al. 2019[20]) refer to the term *environment*. In both cases, the context supports our claim that *environment,* as an element of the generic definition, describes the immediate environment of a human agent engaged in relationship with technology in a workspace.

Table 2. Examples of precise statements based on the generic definition of *sociotechnical system.*

Relevant quotations	**Publication**
"sociotechnical systems: systems that involve both complex physical–technical systems and networks of interdependent actors"	de Bruijn et al. (2009)[22]
"Socio-technical systems (STS) is a theory in which there is an acknowledgement that technology constitutes both social and technical parts, both of which work together to find solutions to problems (Coiera 2007; Mumford 2000, 2006; Appelbaum 1997)".[23-26]	Organ and Stapleton (2015)[27]
"The term socio-technical systems was originally coined by Emery and Trist (1960)[28] to describe systems that involve a complex interaction between humans, machines and the environmental aspects of the work system."	Baxter et al. (2010)[29]

"A socio-technical system is a collection of stakeholders, their networks, practices, and knowledge; the technologies they use; their collective representations; and the standards and rules they adopt."	Yang et al (2021)[10]
"By sociotechnical we mean systems that comprise social elements (namely, humans) as well as technical elements (such as pumps, valves, reactors, etc.). The human elements are an integral part of the system and are often the cause of major systemic failures."	Venkatasub-ramanian et al. (2018)[30]
"The term sociotechnical includes the interaction between social and technical factors when describing conditions underlying system performance...at three levels; the primary work system, the whole organisation and the macrosocial phenomena.[31]"	Njå et al. (2017)[32]
"A sociotechnical system (STS) consists of complex interaction between social humans and technical systems. [33]"	Oosthuizen and Pretorius (2016)[14]
"Maier and Rechtin [2002] offer the following definition of the class of socio-technical systems—technical works involving significant social participation, interest, and concern. [34]"	Haskins (2007)[35]
"'socio-technical' systems are of a hybrid nature because they are constituted by different kinds of elements, intentional and nonintentional: social institutions, human agents and technical artefacts."	Kroes et al. (2006)[36]
"Socio-technical systems (STS) are the building blocks of everyday life. The term has traditionally been used to identify technology in its social settings, but now it identifies systems that comprise people, technologies, physical surroundings, processes and information.[37] Examples include transportation systems, military organizations and corporate enterprises."	Shah et al. (2005)[38]
"Within sociotechnical systems, a population of independent agents (often people) interacts with each other, the technical artifacts in each system, and the environment.[39]"	Watson et al. (2020)[40]
"The term "socio-technical system" (STS) is used to define systems that rely on a 'collaboration of humans' to fulfill its mission.[25] In other words, we use the term STS to represent systems with a social subsystem that performs through collaborative efforts of its constituents and governed by organizational rules. Organizational rules in this context are used	Topcu et al.(2019)[41]

to define where the control boundaries are drawn and how communication/collaboration is coordinated."	

Group 3

Fifteen papers defined terms precisely following the generic definition: "sociotechnical systems involve humans interacting with technology." We list these examples in order of increasing definitional complexity in table form so you can see the way the definitions appear.

Within this third set of papers that formally define the term *sociotechnical system*, a much smaller set of definitions added additional levels of complexity or depth to the generic definition. Both Shah et al. (2005) and Watson et al. (2020) bring in the concept of environment, though they do not define the concept fully.[38, 40] Topcu et al. (2019) includes the concept of organizational rules that govern the interactions of the agents involved in the sociotechnical system.

Within the third group of papers, there were a few that added very specific elements to the generic definition. We quote these at some length to illustrate the levels of complexity currently available to systems engineers within the term *sociotechnical system.*

Table 3. Complex definitions of *sociotechnical system.*

Relevant quotations	**Publication**
"In general, sociotechnical systems such as transportation systems, energy systems, and water supply systems have a scale that is smaller compared to socioecological systems—both in terms of spatial and temporal dimensions. Socioecological systems are deeply linked to the nature and structures of extremely long durations—a much larger spatiotemporal scale and polyrhythmic and dynamic behavior. These systems are marked by physical and natural forces ('acts of God') that oftentimes are beyond the explicit control of people involved in those systems. In contrast, sociotechnical systems are intentional hybrids (people, organizations, society, and technologies—all intertwined); they are constitutive of man-machine interactions that often drive the system toward precarious states."	Amir et al. (2018)[42]
"The term socio-technical systems is nowadays widely used to describe many complex systems, but there are five key	Baxter et al. (2011)[29]

characteristics of open socio-technical systems (Badham et al., 2000): Systems should have interdependent parts. Systems should adapt to and pursue goals in external environments. Systems have an internal environment comprising separate but interdependent technical and social subsystems. Systems have equifinality. In other words, systems goals can be achieved by more than one means. This implies that there are design choices to be made during system development. System performance relies on the joint optimisation of the technical and social subsystems. Focusing on one of these systems to the exclusion of the other is likely to lead to degraded system performance and utility.[43]"	
"Complex sociotechnical systems consist of mutual interactions between agents, entities (technical artifacts), and the system environment. Through emergence, their interactions result in properties measurable at the system level. These complex system properties also interact as Energy, Material, Money, and Information (EMMI) flow through their network. The combination of these interactions with the local environment results in SoS emergent characteristics (e.g., resilience). As an "open system," the SoS then undergoes another layer of interaction between itself, the global environment, and inputs to the SoS. At every hierarchical scale, complexity is present and resistant to reductionist engineering approaches."	Watson et al. (2020)[40]
"In complex sociotechnical systems, cognitive and social humans use technology to make sense of situations when making decisions...A sociotechnical system (STS) consists of complex interaction between social humans and technical systems".[33] ....The social subsystem consists of the organisational structure, which encompasses authority structures, reward systems, knowledge, skills, attitudes, values, and needs. The human (socio) and technical interaction can lead to unexpected, uncontrolled, unpredictable, and complex relationships. Because the STS is an open system and is susceptible to external inputs, it will be affected by the complex operating environment."	Oosthuizen and Pretorius (2016)[14]

"Complex sociotechnical systems have a very large number of inter-connected components with nonlinear interactions that can lead to 'emergent' behavior—that is, the behavior of the whole is more than the sum of its parts." AND "Thus, the essential features of a complex sociotechnical system may be summarized as: (1) goal-driven behavior, (2) many agents or components/sub-components, (3) organized in a multi-layered hierarchy or network, (4) nonlinear dynamical interactions among its agents (or components) and with the environment, (5) feedback loops, (6) decentralized control (i.e., local decision making), and (7) emergent behavior."	Venkatasub-ramanian et al. (2016)[44]
"Sociotechnical systems are systems that involve complex interactions between people, organizations, institutions, and technologies.[45-47, 29] Originally coined by Trist to emphasize people and operations in the coal mining industry,[31] this term has subsequently grown in scope and is increasingly used to characterize systems in many technological sectors such as transportation, power, and information, among many others.[29] In proposing a basis for sociotechnical resilience, it is important to note that sociotechnical entities are not simply aggregations of people and technologies; rather, they are intentional hybrids.[48, 36, 49] They are mutually entangled entities that have to be studied as spatiotemporal functional and adaptive wholes. They are both social and technical at the same time—the people and technologies in these systems are social constructs; in turn, these social constructs are supported by the technical dimension of correct functioning and malfunction of the system.[50, 51] Human knowing and acting in these systems involves understanding the context of work that is conducted at various levels of meanings ranging from the physical to the symbolic. As a result, to comprehend the complexity of meanings, contextual activity, and situated decision making, the system has to be observed at various levels of functional abstraction along with the traditional view of structural decomposition.[52, 53] Apart from these various aspects at the level of meanings (individual and shared), there are systemic properties discernible only at the supra-systemic level. In many accident investigations, it has been observed that certain patterns of systemic behavior are emergent and are not	Amir et al. (2018)[42]

reducible to behavior of individual parts, or ongoing local events.”	
“When trying to design models of socio-technical systems, the designer has to deal with many aspects related to the specific characteristics of these types of systems: Interactions: different interactions among social elements (for example, businesses, customers, governmental institutions, and regulatory compliance) and technical systems (for example, manufacturing factories, and utility plants) arise in socio-technical systems; the interactions among customers and businesses on a technical basis support the idea of customer-centred design of services.[54, 55] Big data: socio-technical systems are complex systems in which big data is involved, in volume, diversity of sources, variety, velocity and veracity.[56] Resource access: according to the service dominant logic,[57] socio-technical systems can be also approached as dynamic configurations of resources where value co-creation processes can be highlighted.[58, 59] Competition and cooperation: the complex nature of interactions arising when stakeholders access resources in service systems is central to the development of effective policies for resource management[60, 61]; for socio-technical systems recent research directions show how human societies form self-governing institutions, in which actors compete and cooperate in an attempt to self-regulate the provision and appropriation of resources according to mutually-agreed, conventional rules.[62, 63] Openness: from an information management perspective, socio-technical systems can be approached as open systems,[64] as they have to deal with diverse sources of information embedded in their environment. Changeability: socio-technical systems are designed, and they are evolving in changing environments through different types of interactions among social and technical elements. Adaptability: to be effective and sustainable, activities and interactions evolving within a socio-technical system should adapt to the changing requirements of their environment. Empowerment: different interactions within current socio-technical systems are initiated by empowered customers using different information channels to make smart decisions and interact with companies (for example,	Dragoicea et al. (2015)[68]

service providers) through various communication channels, hoping to get a superior customer experience.[65, 66]"	

From the examples in Table 3, we can see that there are several additional elements that systems engineers can bring to their generic understanding that a sociotechnical system involves humans interacting with technology. Amir et al (2018) adds two concepts: first, they place the generic sociotechnical system into a socioecological matrix, and, second, they make explicit that the sociotechnical/socioecological system is embedded in a spatiotemporal scale that is in a different order of magnitude to the sociotechnical system considered alone.[42] Baxter et al (2011) also includes the external environment as a major factor and adds the concept of goal driven behavior to the generic definition.[29] In their view sociotechnical systems are themselves goal driven entities, and those goals adapt over time. Watson et al. (2020) also include the external environment.[40] They seem to follow Baxter et al (2011) in understanding that the sociotechnical system will interact with that external environment in complex ways based on goals. They add to the generic definition the concept of emergence, that the interaction between the sociotechnical system and the external environment will create a system of systems with emergent behavior. They follow this up with another additional definition element, hierarchical scale. As the system of systems goes through its behavior cycle, many layers of interaction can build up in ways that can form a hierarchy of scale. Oosthuizen and Pretorius (2016) echo the above statements.[14] Their discussion of reward systems implies acceptance that sociotechnical systems involve goal driven behavior based on values and needs; they include the external environment explicitly, defining the sociotechnical system as open; and they see the interaction between the sociotechnical system and that external environment as complex and likely to lead to emergent (unpredictable) behavior structures. Venkatasubramanian et al (2016) explicitly includes the concepts (already mentioned) of external environment, goal-driven behavior, hierarchical scale, emergent behavior and clarifies that the interactions between the many elements of these systems are dynamic, non-linear structures with feedback loops.[44] Amir et al. (2018) adds to the generic definition that sociotechnical systems become "hybrid agents" through the interactions of their constituent parts.[42] These hybrids of people and technologies are "mutually entangled entities that have to be studied as spatiotemporal functional and adaptive wholes." In other words, the sociotechnical system is an agent operating with other agents in a system of systems. Dragoicea et al. (2015) echoes many of the concepts stated above and adds additional complexity in the following way. They see competition between the agents in sociotechnical systems directly leading to self-regulation that can have emergent properties as they systems adapt over time. This process can lead to evolutionary transformation.[68]

We conclude from this analysis that there is a generic definition of the term *sociotechnical system* within the systems engineering community. That definition as provided above is: "Sociotechnical systems involve humans interacting with technology". However, as engineers explore this concept in their work, additional levels of complexity are being added to the generic concept. We will discuss this further in our conclusions section.

Our findings from this literature review have been summarized in a concept map (see Figure 2). The definition of *sociotechnical system* has origins in systems science for most systems engineering documents or sociotechnical systems theory for a sub-set of documents from the ergonomics and human systems integration systems engineering specialties. These two origin theories are used in the concept map to structure the concepts. The shade of green indicates the frequency of the concept in the systems science-based definitions of *sociotechnical system*. The darker the green the more frequently that concept was used in the definition of *sociotechnical system*.

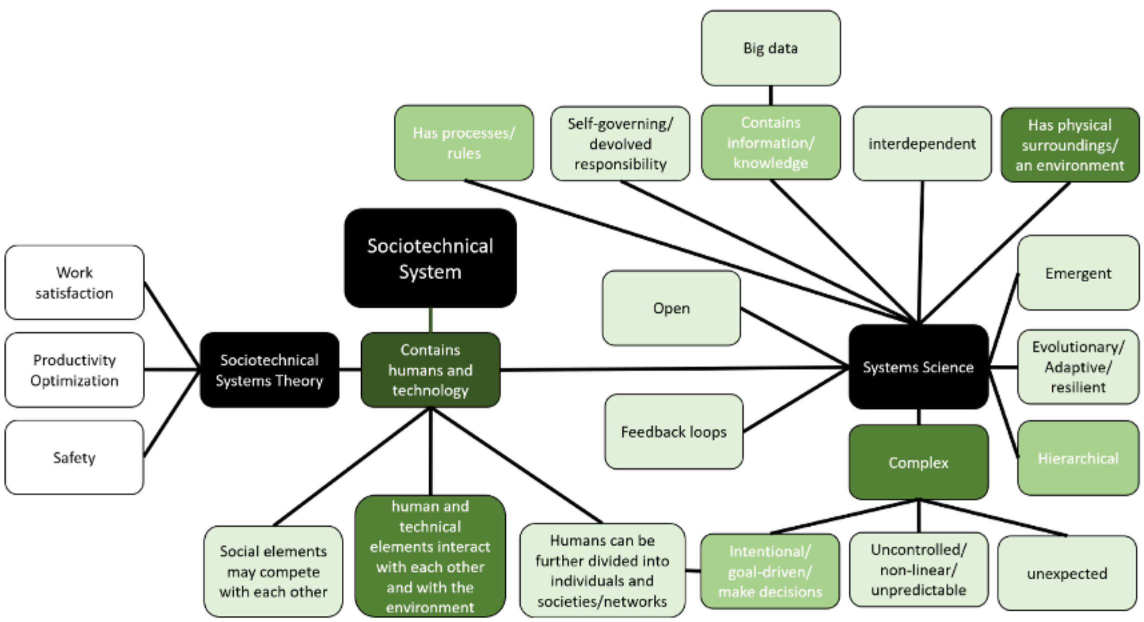

Figure 2. Sociotechnical System Concept Map from https://doi.org/10.1002/sys.21664)

Sociotechnical Systems Theory vs. Sociotechnical System in Systems Engineering
(based on literature review https://doi.org/10.1002/sys.21664)

From the evaluation of all the included papers that discuss and/or analyze the term *sociotechnical system* in a systems engineering context, two theoretical points of departure emerged. The overall set of papers came from many application areas. From that overall set of papers, a smaller set of papers came from human factors and safety with a subgroup in ergonomics. The ergonomics papers and some (but not all) of the papers related more generally to human factors and safety come from the sociotechnical systems theory approach.

While the name "sociotechnical systems theory" sounds general, the theory is specific to ergonomics. The literature review in this study is not a comprehensive theoretical evaluation of sociotechnical systems theories that are present in many different disciplines. However, the sociotechnical systems theory present in systems engineering literature requires some background. Within ergonomics and more generally research on human factors and safety, sociotechnical systems theory is rooted in the work of Rasmussen.[8, 9] While sociotechnical systems theory is well-established in these areas, this does not necessarily mean that this theory is used in conjunction with systems engineering methods. What we see in the results of this systematic literature review is the overlap of research that uses sociotechnical systems theory in ergonomics and/or that uses a systems engineering approach. In ergonomics, sociotechnical systems theory regards the optimization of productivity, worker satisfaction and safety between technology and people in organizations.[69] There is also a strong focus on labor organization in relation to technology.[70]

Sociotechnical systems theory in the tradition of Rasmussen is not the same theoretical position that most of the papers took in this systematic literature review. There were four papers that specifically mentioned Rasmussen's work (papers 41,42, 81, 82 in Table A.1), which either gave definitions of sociotechnical systems that cited Rasmussen's work (paper 81) or gave Rasmussen citations, but gave no explicit definition (paper 41, 42, 82). There were 18 papers from ergonomics (Papers 4-6, 12, 16, 25, 27, 28, 34, 39-42, 50, 55, 64, 81, 82), and those not mentioning Rasmussen gave definitions that were domain specific, such as the Systems Engineering Initiative for Patient Safety (SEIPS) papers (papers 5, 6, 12, 16, 24, 27, 39, 55).

These SEIPS papers are rooted in Holden et al. (2013),[71] which is detailed in Wooldridge et al. (2017).[72] Carayon et al. (2020) provides the most comprehensive definition of sociotechnical systems of all the SEIPs papers in this literature review, including the overall framework for sociotechnical systems in healthcare.[73] Generally however, the SEIPS papers have a working definition of sociotechnical systems: a system with people interacting in an organizational system that involves technology (in these papers, healthcare technology).

In the papers outside of ergonomics, the other theoretical point of departure is Kroes et al. (2006),[36] which is rooted in systems science and engineering. Kroes et al. (2006) is one of the papers in the literature review results as it is specific to systems engineering (Paper 77). Their paper explores engineering systems as sociotechnical systems on a theoretical level. Although many of the papers included in this review do not give an explicit definition of sociotechnical systems, and excepting the 18 ergonomics papers, the use of term sociotechnical systems falls under the Kroes et al. (2006) definition in their implied meaning.

Kroes et al. (2006) explain that engineered systems as technical artifacts include human agents in many parts of the lifecycle of the artifact. This includes not only those in engineering and those using the artifacts, but also the social institutions and their artifacts (e.g., policy, regulation and law). The social and the technical elements are all woven together to make up sociotechnical systems, i.e., all engineering systems are sociotechnical systems. This is a much wider interpretation and has a different character than the definition of a sociotechnical system in the Rasmussen tradition.

Important to note is that many papers use the term *complex sociotechnical systems*. There is no difference between the working definition of sociotechnical systems and complex sociotechnical systems. Though there were papers that directly addressed issues of complexity in sociotechnical systems as discussed above, none of the papers addressed the point that sociotechnical systems might not be complex.

- Situatedness of sociotechnical systems within an external environment
  - Open nature of sociotechnical systems
  - There are hierarchies of scale within the system of systems of which sociotechnical systems are a part (for example, sociotechnical systems within socioecological systems)
  - Sociotechnical systems interact with other sociotechnical systems
- Sociotechnical systems are complex systems
  - They can involve goal driven behavior
  - Interactions between elements of these systems lead to emergent behavior (behavior that is unpredictable, uncontrolled)
  - Interactions between agents and elements of these systems can be:
    - Adaptive
    - Evolutionary
- Agents have a variety of roles and qualities in sociotechnical systems:
  - They self-organize in response to system rules
  - They have goals
  - They are adaptive
  - They can be empowered
  - They can be hybrid in nature (serving multiple functions)
  - They can compete
- Big data, which is defined by Dragoicea et al.[68]: "socio-technical systems are complex systems in which big data is involved, in volume, diversity of sources, variety, velocity and veracity.[56]
- Resource access, which is defined by Dragoicea et al.[68]: "according to the service dominant logic,[57] socio-technical systems can be also approached as dynamic configurations of resources where value co-creation processes can be highlighted.[58, 59]

III. Working within Sociotechnical Systems: Case Studies

Initial thoughts are available in https://doi.org/10.1002/sys.21664 The literature review contains preliminary cases from which we derived definitions, but the document could use fully worked examples.

IV. Policy Modeling for Sociotechnical Systems

V. Accepting additional topics for the primer